

M1S2 | OOP

Class is ...

an object's draft which defines its
attributes and methods

OOP Principles

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data and methods related to an object are combined in one place (class)

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Abstraction

allows you to highlight the general characteristics and functionality of objects or systems, ignoring the implementation details

OOP Pros & Cons

✓ **Modularity and Code Reuse**

allows you to create classes that can serve as reusable modules

✓ **Improved Separation of Concerns**

makes it easy to identify which part of the code is responsible for a specific functionality

✗ **Tendency to Be Confusing**

class hierarchies often lead to multiple method overridings

Lang Construction

Dunder(Magic) Methods

`__init__`, `__str__`, `__repr__`, `__len__`, `__getitem__`,
`__iter__`, ...

```
main.py x
1 class Person: 1 usage
2     ...
3
4 print(dir(Person))
```

Run main x

/Users/a89088/PycharmProjects/PythonProject/.venv/bin/python /Users/a89088/PythonProject/main.py
['__class__', '__delattr__', '__dict__', '__dir__', '__doc__', '__eq__',
Process finished with exit code 0

MRO(Method Resolution Order)

Multiple Inheritance

algorithm for searching methods and attributes in inheritance hierarchy

```
main.py x
1 class A: 2 usages
2     def method(self):
3         return "A's method"
4
5 class B(A): 1 usage
6     def method(self): 1 usage
7         return "B's method"
8
9 class C(A): 1 usage
10    def method(self):
11        return "C's method"
12
13 class D(B, C): 2 usages
14     ...
15
16 print(D.__mro__)
17
18 obj = D()
19 print(obj.method())
```

```
Run main x
/Users/a89088/PycharmProjects/PythonProject/.venv/bin/python /Users/a89088/PycharmProjects/PythonProject/main.py
(<class '__main__.D'>, <class '__main__.B'>, <class '__main__.C'>, <class '__main__.A'>, <class 'object'>)
B's method
Process finished with exit code 0
```



Single Responsibility

module must be responsible for one and only one actor

Open-Closed

software entities should be open for extension, but closed for modification

Liskov Substitution

functions that use a base type should be able to use subtypes of the base type without knowing about it

Interface Segregation

multiple client-specific interfaces are better than one general-purpose interface

Dependency Inversion

dependency on Abstractions. No dependency on implementation